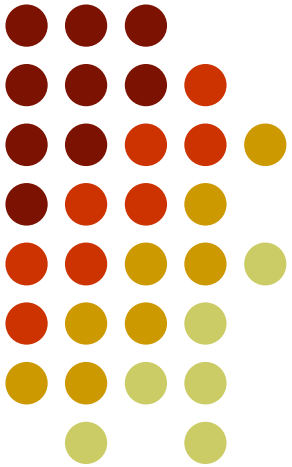
Determination of Forward and Futures Prices

Chapter 5



Consumption vs Investment Assets



- Investment assets are assets held by significant numbers of people purely for investment purposes (Examples: gold, silver)
- Consumption assets are assets held primarily for consumption (Examples: copper, oil)

Short Selling



- Short selling involves selling securities you do not own
- Your broker borrows the securities from another client and sells them in the market in the usual way

Short Selling (continued)

- At some stage you must buy the securities so they can be replaced in the account of the client
- You must pay dividends and other benefits that the owner of the securities receives
- There may be a small fee for borrowing the securities

Example

- You short 100 shares when the price is $100 and close out the short position three months later when the price is $90
- During the three months a dividend of $3 per share is paid
- What is your profit?
- What would be your loss if you had bought 100 shares?

Notation for Valuing Futures and Forward Contracts



S_0: Spot price today

F_0: Futures or forward price today

T: Time until delivery date

r: Risk-free interest rate for maturity T

An Arbitrage Opportunity?

- Suppose that:
  - The spot price of a non-dividend-paying stock is $40
  - The 3-month forward price is $43
  - The 3-month US$ interest rate is 5% per annum
- Is there an arbitrage opportunity?

Another Arbitrage Opportunity?

- Suppose that:
  - The spot price of non-dividend-paying stock is $40
  - The 3-month forward price is US$39
  - The 1-year US$ interest rate is 5% per annum
- Is there an arbitrage opportunity?

The Forward Price

If the spot price of an investment asset (that provides no income) is S_0 and the futures price for a contract deliverable in T years is F_0, then

$$F_0 = S_0 e^{rT}$$

where r is the T-year risk-free rate of interest.

In our examples, S_0 =40, T=0.25, and r=0.05 so that

$$F_0 = 40e^{0.25\times 0.05} = 40.5$$

When Interest Rates are Measured with Continuous Compounding



$$F_0 = S_0 e^{rT}$$

This equation relates the forward price and the spot price for any investment asset that provides no income and has no storage costs

If Short Sales Are Not Possible..

Formula still works for an investment asset because investors who hold the asset will sell it and buy forward contracts when the forward price is too low

When an Investment Asset Provides a Known Income

$$F_0 = (S_0 - I\,)e^{rT}$$

where I is the present value of the income during life of forward contract

Example: Known Income



- Consider a forward contract to purchase a coupon-bearing bond whose current price is $900. The forward contract matures in nine months. A coupon payment of $40 is expected after four months. Assume the four-month and nine-month risk-free interest rates continuously compounded are 3% and 4% per annum, respectively. Consider the following two cases.
  a. The forward price is $910.
  b. The forward price is $870.

When an Investment Asset Provides a Known Yield

$$F_0 = S_0 e^{(r-q)T}$$

where q is the average yield during the life of the contract (expressed with continuous compounding)

Example: Known Yield



- Consider a six-month forward contract on an asset that is expected to provide income equal to 2% of the asset price once during a six-month period. The risk-free rate of interest with continuous compounding is 10% per annum. The asset price is $25.

Valuing a Forward Contract



- A forward contract is worth zero when it is first negotiated
- Later it may have a positive or negative value
- Suppose that K is the delivery price and F_0 is the forward price for a contract that would be negotiated today

Valuing a Forward Contract



- By considering the difference between a contract with delivery price K and a contract with delivery price F_0 we can deduce that:
  - The value, f, of a long forward contract is
    $$(F_0 - K)e^{-rT}$$
  - The value of a short forward contract is
    $$(K - F_0)e^{-rT}$$

Forward vs Futures Prices



- When the maturity and asset price are the same, forward and futures prices are usually assumed to be equal. (Eurodollar futures are an exception)
- When interest rates are uncertain they are, in theory, slightly different:
  - A strong positive correlation between interest rates and the asset price implies the futures price is slightly higher than the forward price
  - A strong negative correlation implies the reverse

Stock Index

- Can be viewed as an investment asset paying a dividend yield
- The futures price and spot price relationship is therefore

$$F_0 = S_0 e^{(r-q)T}$$

where q is the dividend yield on the portfolio represented by the index during life of contract

Index Arbitrage



- When $F_0 > S_0 e^{(r-q)T}$ an arbitrageur buys the stocks underlying the index and sells futures
- When $F_0 < S_0 e^{(r-q)T}$ an arbitrageur buys futures and shorts or sells the stocks underlying the index

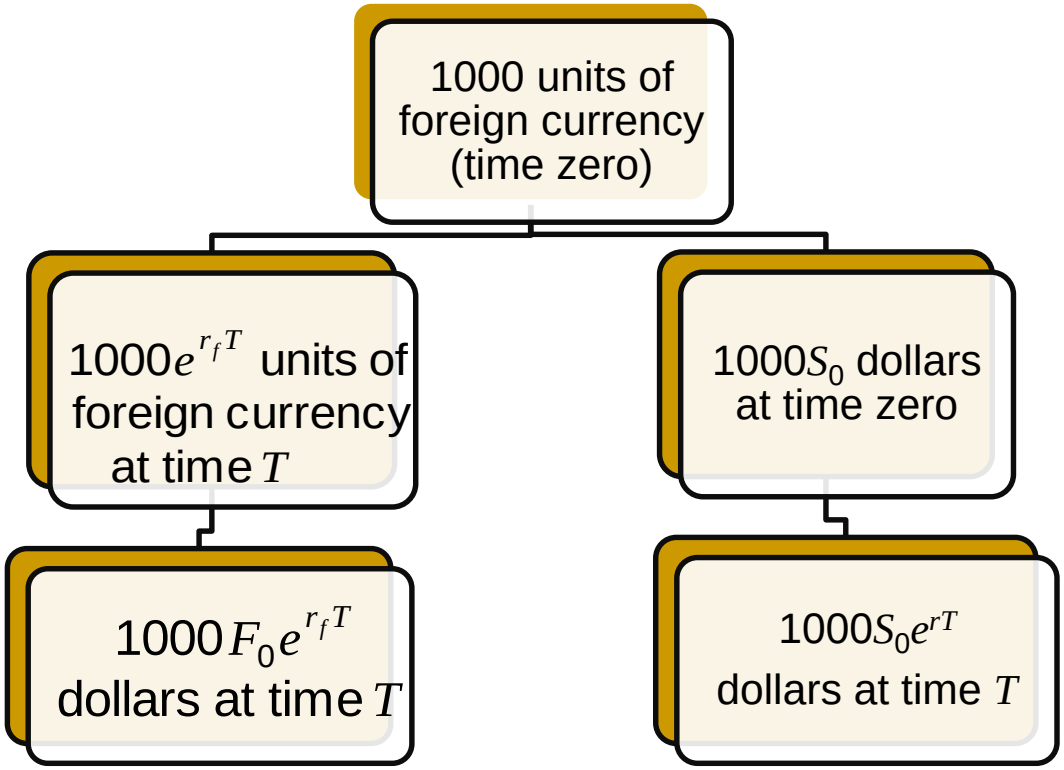
Futures and Forwards on Currencies



- A foreign currency is analogous to a security providing a yield
- The yield is the foreign risk-free interest rate
- It follows that if r_f is the foreign risk-free interest rate

$$F_0 = S_0 e^{(r-r_f)T}$$

Explanation of the Relationship Between Spot and Forward (Figure 5.1, page 118)



Example: Arbitrage Opportunities



- Suppose that the two-year interest rates in Australia and the United States are 5% and 7%, respectively, and the spot exchange rate is 0.6200 USD per AUD. In the following two cases, how could an arbitrageur make a profit?
  a) The two-year forward exchange rate is 0.6300.
  b) The two-year forward exchange rate is 0.6600.

Consumption Assets: Storage is Negative Income

$$F_0 = (S_0 + U)e^{rT}$$

where U is the present value of the storage costs. Alternatively,

$$F_0 = S_0\, e^{(r+u)T}$$

where u is the storage cost per unit time as a percent of the asset value.

Example: Arbitrage Opportunities regarding Gold



- Consider a one-year futures contract on gold. We assume no income and that it costs $2 ounce per year to store gold, with the payment being made at the end of the year. The spot price is $1,600 and the risk-free rate is 5% per annum for all maturities. In the following two cases, how could an arbitrageur make a profit?
  a) The actual gold futures price is $1,700.
  b) The futures price is $1,650.

Consumption Assets: Storage is Negative Income (continued)



- In the case of investment assets such as gold and silver, we can argue that many investors hold the commodity solely for investment.
- However, this argument cannot be used for commodities that are not held for investment.
  - $F_0 \leq (S_0 + U)e^{rT}$
  - $F_0 \leq S_0 e^{(r+u)T}$

The Cost of Carry



- The cost of carry, c, is the storage cost plus the interest costs less the income earned
- For an investment asset $F_0 = S_0 e^{cT}$
- For a consumption asset $F_0 \leq S_0 e^{cT}$

Futures Prices & Expected Future Spot Prices



- Suppose k is the expected return required by investors in an asset
- We can invest $F_0 e^{-r\,T}$ at the risk-free rate and enter into a long futures contract to create a cash inflow of S_T at maturity
- This shows that

$$F_0 e^{-rT} e^{kT} = E(S_T)$$

or

$$F_0 = E(S_T) e^{(r-k)T}$$

Futures Prices & Expected Future Spot Prices (continued)

No Systematic Risk	$k = r$	$F_0 = E(S_T)$
Positive Systematic Risk	$k > r$	$F_0 < E(S_T)$
Negative Systematic Risk	$k < r$	$F_0 > E(S_T)$

Positive systematic risk: stock indices
Negative systematic risk: gold (at least for some periods)

Summary

- Main concepts
  - Investment assets and Consumption assets
  - Short selling
    - Selling securities that you do not own
    - Must pay dividends and other benefits that the owners of the securities receives

Summary

- For investment assets

Table 5.6 Summary of results for a contract with time to maturity T on an investment asset with price S_0 when the risk-free interest rate for a T-year period is r.

Asset	*Forward/futures price*	*Value of long forward contract with delivery price K*
Provides no income:	$S_0 e^{rT}$	$S_0 - Ke^{-rT}$
Provides known income with present value I:	$(S_0 - I)e^{rT}$	$S_0 - I - Ke^{-rT}$
Provides known yield q:	$S_0 e^{(r-q)T}$	$S_0 e^{-qT} - Ke^{-rT}$

Summary

- Two specific futures and forwards
  - Futures/Forwards on stock index
    - Stock index can be viewed as an investment asset paying a dividend yield

$$F_0 = S_0\, e^{(r-q)T}$$

  - Futures/Forwards on currencies
    - A foreign currency is analogous to a security providing a yield

$$F_0 = S_0 e^{(r-r_f)T}$$

Summary

- When storage cost is involved
  - Storage is negative income
  - For investment assets

$$F_0 = (S_0 + U)e^{rT} \qquad F_0 = S_0\, e^{(r+u)T}$$

  - For consumption assets

$$F_0 \le (S_0 + U)e^{rT} \qquad F_0 \le S_0\, e^{(r+u)T}$$

Summary

- Cost of carry
  - c = storage cost + interest costs – income
  - For investment assets

$$F_0 = S_0 e^{cT}$$

  - For consumption assets

$$F_0 \leq S_0 e^{cT}$$